Akoizo Original Practice - Form 4 - Answer key

Division C | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which material is NOT allowed in a Boomilever?

Unit: Wood selection, density, and buckling.
Worked-case data: A strut buckles at 120 N with an unbraced length of 30 cm. A brace is added at its midpoint. Estimate the new buckling load.
Separate investigation observations:
Stick A: 0.11 g/cm^3 - failed as a strut at 6 kg.
Stick B: 0.18 g/cm^3, same size - held 11 kg as a strut.
Stick C: 0.11 g/cm^3 with a midpoint brace - held 15 kg.

A. Basswood

B. Hard balsa

C. Commercial plywood

D. Team-laminated balsa

Correct: C

2. (2 points) How do compression members usually fail in a Boomilever?

Unit: Statics of a cantilever: moments, tension, and compression.
Worked-case data: A 15 kg load hangs 42 cm from the wall. The strut meets the wall 15 cm below the bolt. Estimate the horizontal chord force.
Separate investigation observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. By crushing the bolt

B. By buckling sideways

C. By stretching apart

D. By melting the glue

Correct: B

3. (2 points) A joint failed and glue remained on only one surface. What does that suggest?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

A. Poor bond area or fit at that joint

B. The wood was too dense

C. The load was too far out

D. The bolt was loose

Correct: A

4. (2 points) Why do soft-wood bolt hooks often fail by crushing?

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. The round bolt concentrates force on a small area of soft wood

B. The bolt heats up under load

C. Glue cannot touch metal

D. The bolt is too far from the wall

Correct: A

5. (4 points) Explain why the bonus contact line usually makes a Boomilever heavier.

answer: A shallower contact shortens the lever arm of the couple, so chord and strut forces rise, and a flatter strut buckles more easily, requiring more wood or bracing. (4 points).

6. (2 points) Which prototype should the team bring, based on expected score?

Supplied synthetic observations:
Prototype 1 (base): 9.6 g, held 15,000 g, contact at 14 cm below the bolt.
Prototype 2 (bonus): 12.1 g, held 15,000 g, contact above the 10 cm line.
Prototype 3 (bonus, lightened): 10.4 g, failed at 12,300 g when a joint cracked.
Supervisor note: Prototype 1's contact point is above the 15 cm line.

A. Prototype 3

B. Prototype 2

C. Prototype 1

Correct: B

7. (4 points) Why is pouring sand steadily better than dumping it quickly?

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

answer: Fast pouring creates sudden dynamic loads and bucket swing that can exceed the static load and break the structure early. (4 points).

8. (4 points) Give one reason the tension chord often fails at its ends rather than its middle.

answer: Force must pass through glue joints and the bolt hook, where contact areas are small and stress concentrates, while the middle carries only uniform tension. (4 points).

9. (2 points) What do these tests show about where the structure needs material?

Supplied synthetic observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. All members should be the same size

B. The strut needs bracing and joints need care; the chord is already strong enough

C. The chord needs to be much thicker

Correct: B

10. (4 points) How does humidity affect a finished structure?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

answer: Wood absorbs moisture, adding mass and slightly reducing stiffness, so store structures dry and weigh them before competition. (4 points).

11. (6 points) Solve this case and explain your reasoning: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?

reasoning: Required strength: 300 N x 1.5 = 450 N.
Area = 450 N ÷ 2 N/mm^2 = 225 mm^2.
A 15 mm x 15 mm overlap, or two gussets of about 113 mm^2 each, would do. (3 points).

conclusion: Joint area, not member size, decides whether the chord end survives. (3 points).

12. (2 points) Define “Contact depth” in the context of Boomilever.

answer: Vertical distance from the bolt centerline down to the wall contact. (2 points).

13. (2 points) Define “Loading Assembly” in the context of Boomilever.

answer: Block, eyebolt, chain, S-hook, and bucket; at most 1,500 g. (2 points).

14. (2 points) Define “Testing Wall” in the context of Boomilever.

answer: Plumb, rigid wall at least 40 x 30 cm holding the bolt. (2 points).

15. (2 points) Define “Modulus of elasticity (E)” in the context of Boomilever.

answer: Material stiffness; resistance to stretching or bending. (2 points).

16. (2 points) Define “Pyramid layout” in the context of Boomilever.

answer: Paired chords and struts that resist twisting and side loads. (2 points).

17. (2 points) Define “Free-body diagram” in the context of Boomilever.

answer: A sketch of one object with every external force acting on it. (2 points).

18. (2 points) Define “Safety factor” in the context of Boomilever.

answer: Ratio of capacity to expected load; margin against variation. (2 points).